

Supplementary Material: Machine Learning Discovers a Robust 36-Gene Transcriptomic Signature for Sepsis Mortality

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Abstract

This supplementary document provides extended methodological details, quality control metrics, and comprehensive tabular results supporting the discovery of the 36-gene prognostic signature for sepsis mortality. It includes parameters for data harmonization (Empirical Bayes ComBat), the Recursive Feature Elimination with Cross-Validation (RFECV) pipeline, XGBoost hyperparameter tuning, and Leave-One-Cohort-Out (LOCO) meta-analysis across the 1,636-patient multi-cohort atlas. Full lists of the initial candidate differentially expressed genes (DEGs) and functional enrichment pathways are also provided to ensure complete computational and biological reproducibility.

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1 Data Acquisition and Preprocessing

1.1 Data Retrieval and Cohort Filtering

Sepsis transcriptomic cohorts were identified from the NCBI Gene Expression Omnibus (GEO) using a programmatic search focused on human PBMC and whole-blood datasets. The search query was:

```
sepsis[Title] AND "Homo sapiens"[Organism] AND (blood[Description] OR PBMC[Description]) AND "gse"[Filter]
```

GEO Series records matching the query were retrieved using the NCBI Entrez E-utilities API. For each candidate cohort, accession ID, study title, sample count, and platform information were extracted and compiled into a master index (`geo_sepsis_master_menu.csv`). Cohorts were then screened by sample size, retaining datasets with more than 100 samples for downstream processing.

Data retrieval was handled according to assay type. For microarray studies, compressed GEO family files were downloaded from the corresponding NCBI FTP locations. For RNA-seq studies, which often provide study-specific supplementary count matrices, datasets were flagged for manual retrieval and placed in the RNA-seq raw-data directory after download. This approach allowed microarray and RNA-seq cohorts to be organized consistently while preserving the original data formats used by each study.

After data acquisition, cohort metadata were extracted and reviewed to harmonize clinical labels across studies. Because GEO records often use different terms for patient status, disease severity, and mortality outcomes, microarray and RNA-seq metadata were processed with modality-specific workflows and converted into structured patient-level metadata files. Summary reports of the observed clinical terms were then used to define rule-based mappings into standardized binary survival labels before construction of the machine learning dataset.

1.2 Matrix Parsing and Probe Mapping

Following metadata extraction, expression matrices and clinical labels were parsed together to generate cohort-level files for downstream analysis. A main challenge at this stage was that GEO studies differed in both data format and clinical annotation. The parser was therefore designed to retain the rationale of the study design while standardizing the inputs required for model development.

First, the analysis was restricted to confirmed sepsis samples. Records annotated as controls, healthy subjects, or non-sepsis inflammatory conditions were excluded so that feature selection would focus on mortality-associated variation within sepsis rather than broader differences between disease and non-disease groups. Mortality labels also required harmonization, as cohorts used different terms for survivors and non-survivors. These terms were mapped to a common binary endpoint (1 = death, 0 = survival), and samples without a clear mortality label were removed.

Second, expression matrices were constructed according to assay type. For microarray cohorts, probe-level expression values were extracted from GEO family files. For RNA-seq cohorts, count matrices were obtained from the supplementary files provided by each study, with sample names matched back to GEO sample identifiers when necessary. This step addressed the practical differences between GEO microarray records and study-specific RNA-seq deposits.

Finally, expression matrices were aligned with the harmonized clinical labels. Samples were

retained only when both expression data and a standardized outcome label were available. The resulting matrices and clinical label files were then used for normalization, batch correction, and downstream model development.

A shared gene-level feature space was required before combining cohorts across platforms. The raw matrices used different identifier systems: microarray datasets were reported at the probe level, whereas RNA-seq datasets often used Ensembl identifiers. These identifiers were therefore mapped to standard HUGO gene symbols before cohort integration.

For microarray datasets, platform annotation files from GEO were used to link probe identifiers to gene symbols. Because platform files vary in structure and naming conventions, the mapping step checked the available annotation fields and extracted valid gene symbols where possible. When direct gene-symbol annotations were incomplete, Entrez-based mappings were used as a secondary source. For RNA-seq datasets, Ensembl identifiers were converted to HUGO gene symbols after removing version suffixes when present.

After identifier conversion, some genes were represented by multiple probes or repeated mappings. These duplicate gene-level entries were collapsed by averaging expression values across mapped probes or rows. Unmapped identifiers were removed. This produced one standardized gene-level matrix per cohort, allowing the subsequent normalization and batch-correction steps to operate on a common feature space.

1.3 Normalization and Batch Correction

The integration of 1,636 transcriptomic profiles across seven independent cohorts introduced technical variation from differences in sample processing, assay type, sequencing depth, and microarray platform design. To reduce these effects while preserving the mortality signal, the harmonization workflow used a shared gene space, within-cohort standardization, Empirical Bayes batch correction, and a separate external validation strategy.

1.3.1 Feature Space Intersection

Before merging cohorts, all expression matrices were restricted to genes present in every dataset. This step created a common coordinate system for machine learning and avoided introducing missing values or synthetic imputation for genes absent from one or more cohorts.

Mathematically, if G_i represents the set of mapped HUGO gene symbols in cohort i , the shared background transcriptome G_{core} was defined as:

$$G_{core} = \bigcap_{i=1}^N G_i$$

For example, if three cohorts contained the following gene sets:

- Cohort A: TP53, IL6, S100P, TNF
- Cohort B: TP53, IL6, S100P, CXCL8
- Cohort C: TP53, IL6, S100P, KLF9

then the retained shared feature space would be TP53, IL6, and S100P. Genes present in only a subset of cohorts were excluded from the merged matrix. In this study, the intersection produced a shared background transcriptome of 7,964 genes.

1.3.2 Within-Cohort Standardization

Raw expression values are not directly comparable across assay types. For example, an RNA-seq count value and a microarray fluorescence value are measured on different scales, even when they refer to the same gene. To place samples on a comparable scale, Z-score standardization was applied within each cohort before merging.

For gene g , patient p , and cohort c , the standardized expression value was calculated as:

$$Z_{g,p,c} = \frac{X_{g,p,c} - \mu_{g,c}}{\sigma_{g,c}}$$

where $X_{g,p,c}$ is the original expression value, $\mu_{g,c}$ is the mean expression of gene g within cohort c , and $\sigma_{g,c}$ is the corresponding within-cohort standard deviation.

Within-cohort standardization transforms expression values into relative deviations from each cohort-specific baseline, allowing measurements originating from different assay platforms to be compared on a common scale. This step reduces scale-related technical variability prior to cross-cohort harmonization.

1.3.3 Empirical Bayes Batch Correction

Within-cohort Z-scoring standardizes the scale of each dataset, but it does not fully remove systematic technical variation arising from differences in sequencing platforms, sample-processing protocols, and cohort-specific effects. To reduce these residual sources of variation, Empirical Bayes batch correction was performed using the `neuroComBat` Python implementation of the ComBat algorithm. Cohort identity was specified as the batch variable, and mortality status was included as a biological covariate to preserve outcome-associated expression differences during harmonization.

The ComBat model represents standardized expression Y_{ijg} for gene g , sample j , and batch i as:

$$Y_{ijg} = \alpha_g + X_j\beta_g + \gamma_{ig} + \delta_{ig}\epsilon_{ijg}$$

where α_g denotes the overall baseline expression of gene g , $X_j\beta_g$ represents the biological covariate effect, γ_{ig} is the additive batch effect, δ_{ig} is the multiplicative batch effect, and ϵ_{ijg} is the residual error term.

In practice, ComBat estimates cohort-specific additive and multiplicative effects using a shared Empirical Bayes framework and adjusts expression values to reduce technical heterogeneity while preserving biologically meaningful variation. Inclusion of mortality status in the design matrix helps prevent harmonization from removing expression differences associated with survival outcome.

Importantly, ComBat correction was applied only to datasets used for training and exploratory analyses. The external validation cohort (GSE65682) was excluded from the harmonization procedure and retained as an independently standardized holdout dataset to prevent information leakage and provide a stricter assessment of model generalizability.

1.3.4 Training, Visualization, and External Validation Matrices

To reduce the risk of data leakage, separate matrices were prepared for visualization, model development, and external validation. This separation was important because batch correction

estimates global parameters from the datasets provided to it; including a held-out test cohort during correction could indirectly expose the training workflow to information from the validation data.

Three matrices were therefore maintained:

- **Atlas matrix:** all seven cohorts were harmonized together for descriptive analyses and visualization, such as PCA plots. This matrix was not used for model fitting.
- **Training matrix:** six cohorts were harmonized together and used for differential expression analysis, feature selection, cross-validation, and hyperparameter tuning.
- **External validation matrix:** GSE65682 was kept separate from the ComBat-corrected training matrix and was processed independently after within-cohort standardization. This provided a more realistic assessment of model performance on an unseen cohort.

This design allowed the study to use batch-corrected data for model development while keeping the external validation cohort isolated from the training-stage harmonization process.

2 Differential Expression and Feature Selection

2.1 Cross-Cohort DEG Identification

Following data harmonization, the machine learning training matrix contained 7,964 intersecting genes. Training a classifier directly on this full feature space would be inefficient and could increase the risk of overfitting, particularly because many genes may not carry consistent mortality-related information. To reduce the feature space while retaining biologically interpretable signals, differential expression analysis was used as an initial filtering step before RFECV-based machine learning feature selection.

2.1.1 Within-Cohort Welch’s T-Test

Differential expression was first assessed independently within each of the six training cohorts. Welch’s t-test was used instead of the standard Student’s t-test because it does not assume equal variance between the two outcome groups. This was appropriate for sepsis transcriptomic data, where non-survivors and survivors may differ not only in mean expression but also in expression variability.

For each gene, the Welch’s t-statistic was calculated as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

where $\bar{X}_1$ and $\bar{X}_2$ are the mean Z-scored expression values for non-survivors and survivors, respectively; s_1^2 and s_2^2 are the corresponding sample variances; and N_1 and N_2 are the sample sizes of the two groups.

As an example, consider the gene S100P in one cohort. Suppose non-survivors ($N_1 = 30$) have a mean Z-score of 1.8 with variance 1.2, while survivors ($N_2 = 100$) have a mean Z-score of -0.2 with variance 0.4. Because both the variances and group sizes differ, Welch’s t-test provides a more suitable test statistic than a pooled-variance t-test. This helps avoid assuming that the survivor and non-survivor groups have identical expression variability.

2.1.2 False Discovery Rate Correction

Because the analysis tested thousands of genes, raw p-values were adjusted to control the false discovery rate. Without correction, applying a nominal $p < 0.05$ threshold across 7,964 genes would be expected to produce many false positives by chance. The Benjamini-Hochberg procedure was therefore used to convert raw p-values into FDR-adjusted values.

The procedure ranks the raw p-values from smallest to largest and identifies the largest rank k satisfying:

$$P_{(k)} \leq \frac{k}{m}\alpha$$

where m is the total number of tested genes and α is the target false discovery rate, set here to 0.05. Genes with adjusted FDR values below 0.05 were retained for the next filtering stage.

The Benjamini-Hochberg procedure controls the expected proportion of false discoveries among the genes declared significant, providing a less conservative alternative to family-wise error rate corrections while remaining appropriate for genome-scale testing.

2.1.3 Cross-Cohort Consistency Filtering

Statistical significance within a single cohort was not considered sufficient for inclusion. A gene may appear significant in one dataset because of cohort-specific factors, such as differences in clinical composition, treatment setting, pathogen distribution, or platform-specific effects. To prioritize signals that were more consistent across studies, the direction of expression change was evaluated across the six training cohorts.

For each significant gene, the mean expression difference was calculated as:

$$\Delta\mu = \bar{X}_{non-survivor} - \bar{X}_{survivor}$$

Genes were retained only if the direction of change was consistent in at least five of the six training cohorts.

For example, a gene with directional shifts $[+, +, +, +, +, -]$ would pass this filter because it is upregulated in five cohorts and downregulated in only one. In contrast, a gene with shifts $[+, +, +, -, -, -]$ would be excluded because its direction is evenly split across cohorts, making its mortality-associated pattern less stable across datasets.

2.1.4 Composite Scoring and Percentile Truncation

After consistency filtering, the remaining genes were ranked using a composite score that combined effect size and statistical confidence. This score rewarded genes with larger mean differences and stronger adjusted significance:

$$C_g = |\Delta\mu_g| \times (-\log_{10}(\text{FDR}_g))$$

where $|\Delta\mu_g|$ represents the absolute mean expression difference for gene g , and $-\log_{10}(\text{FDR}_g)$ increases as statistical confidence improves.

As a simple example, if Gene X has an absolute mean difference of 1.5 and an FDR of 0.001 (10^{-3}), then:

$$C_g = 1.5 \times (-\log_{10}(10^{-3})) = 1.5 \times 3 = 4.5$$

If the 95th percentile cutoff in its direction category is 3.2, Gene X would pass this ranking threshold and advance to the RFECV stage.

To avoid selecting only strongly upregulated inflammatory genes, upregulated and downregulated genes were ranked separately. Genes were divided into two groups based on the sign of $\Delta\mu$: upregulated genes ($\Delta\mu > 0$) and downregulated genes ($\Delta\mu < 0$). Within each group, only genes above the 95th percentile of the composite score distribution were retained. This preserved high-confidence candidates from both expression directions and produced the final DEG subset used for downstream machine learning feature selection.

2.2 Recursive Feature Elimination (RFECV)

The differential expression pipeline reduced the transcriptome from 7,964 shared background genes to a smaller pool of high-confidence candidates. However, this candidate set was still larger than would be ideal for a practical prognostic assay. The next step was therefore to identify a compact gene panel that retained predictive performance while removing redundant or weakly informative features.

To do this, the candidate genes were passed into an XGBoost-based Recursive Feature Elimination with Cross-Validation (RFECV) framework. The purpose of this step was to evaluate how model performance changed as genes were removed one at a time, and to identify the smallest feature set that maintained the strongest cross-validated discrimination.

2.2.1 Dynamic Cost-Sensitive Weighting

Because mortality was the minority outcome, the classifier needed to account for class imbalance during training. If class imbalance is ignored, a model can achieve apparently high accuracy by favoring the majority survivor class, while performing poorly for non-survivors. To reduce this issue, cost-sensitive learning was used within XGBoost.

The positive class weight was calculated as the ratio of survivor samples to non-survivor samples:

$$w_{pos} = \frac{N_{negative}}{N_{positive}}$$

where $N_{negative}$ represents the number of survivor samples and $N_{positive}$ represents the number of non-survivor samples.

For example, if the training matrix contains 1,000 survivors and 200 non-survivors, then:

$$w_{pos} = \frac{1000}{200} = 5.0$$

In this case, misclassifying a non-survivor receives a higher penalty than misclassifying a survivor. This encourages the model to pay greater attention to mortality-associated patterns rather than defaulting toward the majority class.

2.2.2 RFECV Mechanism

After configuring the weighted XGBoost classifier, Recursive Feature Elimination with Cross-Validation was used to reduce the candidate gene set. The procedure began with the full candidate pool and repeatedly removed the least informative gene based on the model’s feature-importance ranking.

At each iteration, the model was retrained on the remaining genes and evaluated using stratified five-fold cross-validation. Stratification preserved the survivor/non-survivor ratio in each fold,

and AUROC was used as the primary performance metric. The process continued until the performance profile across different feature-set sizes could be assessed.

A simplified example illustrates the logic:

- Loop 1: the model evaluates 64 genes and obtains an AUROC of 0.78; the least informative gene is removed.
- Loop 2: the model evaluates 63 genes and obtains an AUROC of 0.79; the next least informative gene is removed.
- Later iteration: the model evaluates 36 genes and reaches an AUROC of 0.803.
- Next iteration: the model evaluates 35 genes, but the AUROC decreases to 0.795, suggesting that a useful gene or gene interaction may have been removed.

This iterative procedure allows the model to identify whether performance improves as redundant features are removed, and where further pruning begins to reduce predictive value.

2.2.3 Optimal Panel Selection

The final panel was selected by examining the relationship between the number of retained genes and cross-validated AUROC. The goal was to identify the point at which the model retained strong discrimination without carrying unnecessary features.

In this study, RFECV identified the strongest cross-validated AUROC at 36 genes. This 36-gene subset was therefore selected as the final transcriptomic signature for downstream validation and external testing.

3 Machine Learning Modeling and Benchmarking

3.1 Algorithm Benchmarking and Hyperparameter Tuning

After preprocessing, the transcriptomic profiles had been mapped into a shared, Z-scored, and ComBat-corrected training matrix containing 7,964 gene-level features. At this stage, the goal was not yet to define the final biomarker panel, but to determine which type of classifier was best suited to model the full high-dimensional feature space. Benchmarking was therefore performed before recursive feature reduction, so that the downstream selection step could be built around the most appropriate modeling framework.

3.1.1 Benchmarking Protocol

The benchmarking step compared a range of linear and non-linear classifiers, including Logistic Regression with ElasticNet regularization, Support Vector Machines with linear and radial basis function kernels, Random Forest, and XGBoost. These models were chosen to represent different assumptions about the data: linear separation, margin-based classification, ensemble decision trees, and gradient-boosted tree learning. Each model was evaluated on the same training matrix so that performance differences reflected model behavior rather than differences in input features.

Because survivors outnumbered non-survivors, class imbalance was handled using cost-sensitive learning. This allowed the models to place greater weight on the minority mortality class without generating synthetic transcriptomic profiles. Synthetic oversampling methods such as SMOTE were not used because, in high-dimensional gene-expression data, artificial samples can alter the covariance structure of the original matrix. Instead, class weights were estimated from the observed outcome distribution and applied during model training.

Model performance was assessed using stratified five-fold cross-validation. Stratification preserved the survivor/non-survivor ratio within each fold, reducing the chance that a fold-specific class imbalance would influence the comparison between algorithms.

3.1.2 Benchmarking Results and Interpretation

Among the tested models, XGBoost achieved the highest internal cross-validated performance, with an AUROC of 0.89 and stronger precision and F1-score values than the other classifiers. This result suggested that the gradient-boosting framework was better able to capture the non-linear and high-dimensional structure of the harmonized transcriptomic matrix than the other tested model classes.

However, this internal AUROC was interpreted as a model-selection metric rather than as the final estimate of predictive performance. The training cohorts had already been harmonized together using ComBat, which estimates correction parameters from the pooled training data. As a result, samples assigned to different cross-validation folds were not completely independent with respect to the earlier batch-correction step. This does not invalidate the benchmark as a comparison among algorithms, but it means that the internal score can be higher than performance on a fully unseen cohort.

For this reason, the 0.89 AUROC was used to justify the choice of XGBoost for downstream RFECV and hyperparameter tuning, not to claim final generalizability. The more relevant evaluation of transferability was reserved for the isolated external validation cohort, which was processed separately from the ComBat-corrected training matrix. In that held-out setting, the model achieved an AUROC of 0.66, providing a more conservative estimate of performance on an unseen cohort.

3.2 Rationale for XGBoost Modeling

Before detailing model interpretation and validation, it is important to establish why XGBoost was selected over traditional statistical models such as Logistic Regression or standard Support Vector Machines for this biological application.

3.2.1 Linear Assumptions and Biological Reality

Standard linear models assume that each gene’s contribution to mortality risk is additive and independent. A Logistic Regression model estimates log-odds using a weighted linear combination of gene-expression values:

$$\log\left(\frac{p}{1-p}\right) = w_0 + w_1x_{S100P} + w_2x_{TGFB1} + \cdots + w_nx_n$$

However, biological systems are frequently non-linear and synergistic. For example, a patient may enter a fatal hyper-inflammatory state only when one gene is strongly upregulated while

another is simultaneously downregulated. Linear models cannot capture these threshold effects or interaction patterns unless explicit interaction terms, such as $x_A \times x_B$, are engineered in advance.

3.2.2 XGBoost Additive Framework

XGBoost addresses this limitation by using an additive ensemble of decision trees. Instead of estimating a single linear boundary, the model recursively partitions samples according to expression thresholds and combines many weak learners into a stronger classifier.

At iteration t , XGBoost adds a new tree f_t that targets the residual errors of the previous $t - 1$ trees. The objective combines a loss function L with a regularization term Ω :

$$\text{Obj}^{(t)} = \sum_{i=1}^n L\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t)$$

For efficient optimization, XGBoost uses a second-order Taylor expansion based on the first derivative, or gradient g_i , and the second derivative, or Hessian h_i , of the loss function:

$$\text{Obj}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t)$$

3.2.3 Regularization for High-Dimensional Genomic Data

The primary risk when applying decision-tree models to transcriptomic data is overfitting, particularly when the number of genes P is large relative to the number of patients N . XGBoost mitigates this risk through explicit regularization:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 + \alpha \sum_{j=1}^T |w_j|$$

Here, T is the number of leaves in the tree, w_j is the score assigned to leaf j , γ penalizes unnecessary tree complexity, and λ and α apply L2 and L1 penalties to leaf weights. In practical terms, this prevents the model from assigning excessive importance to small, noisy patient subsets and encourages predictions that reflect broader multi-gene patterns.

3.3 Model Explainability with Consensus SHAP

Ensemble models such as XGBoost can be difficult to interpret directly. To extract biological meaning from the optimized 36-gene panel, the pipeline used SHapley Additive exPlanations (SHAP), which quantify each gene's contribution to individual mortality-risk predictions.

3.3.1 SHAP Mathematics

For a gene i , the SHAP value ϕ_i is computed by comparing model predictions across all subsets of features that exclude i and measuring how predictions change when i is added:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [v(S \cup \{i\}) - v(S)]$$

Here, F is the full set of 36 features, S is a feature subset that excludes gene i , and v is the model prediction function.

3.3.2 Fifty-Run Consensus Method

Because XGBoost includes stochastic elements such as column subsampling and random initialization, feature-importance rankings can vary slightly across individual training runs. To stabilize the reported biomarker rankings, 50 independent XGBoost models were trained using different initialization seeds.

For each run, patient-level SHAP values were extracted and converted into mean absolute SHAP values for each gene. These values were then averaged across all 50 runs to produce a consensus importance table. This procedure prioritized genes whose predictive contribution was stable across repeated model fits rather than dependent on a single stochastic training instance.

3.4 Strict External Generalization

The strongest test of a multi-cohort transcriptomic model is its ability to generalize to completely unseen data generated by a different clinical team and sequencing platform without batch-effect leakage from the training cohorts.

3.4.1 Holdout Vault Protocol

For final validation, the optimized 36-gene XGBoost model was trained exclusively on the harmonized six-cohort training matrix, frozen, and evaluated against the locked `X_vault` tensor from GSE65682.

Importantly, `X_vault` was Z-scored but was not included in ComBat harmonization. The model therefore had to operate under the raw technical variability of a new patient population, providing a stricter assessment of external transferability.

3.4.2 Bootstrap Confidence Intervals

A single AUROC point estimate is insufficient for assessing external validation reliability. A 95% confidence interval was therefore generated using non-parametric bootstrapping.

Let the original vault dataset be D with sample size N . A total of 1,000 bootstrap datasets $D_1^*, D_2^*, \dots, D_{1000}^*$ were generated by sampling N observations from D with replacement. AUROC was calculated for each bootstrap replicate, and the 2.5th and 97.5th percentiles of the resulting distribution were used as the confidence bounds.

3.4.3 External Performance Interpretation

The external validation cohort produced an AUROC of 0.66 with a 95% confidence interval of 0.60–0.72. Although lower than the internal cross-validation performance, this reduction is

expected because the holdout cohort was intentionally excluded from ComBat harmonization, differential expression analysis, feature selection, and model optimization.

Consequently, the external validation protocol represents a substantially stricter assessment of generalizability than standard cross-validation performed within the harmonized training matrix. The observed performance therefore reflects the ability of the 36-gene signature to retain prognostic information under realistic cohort shift, including differences in patient populations, clinical practices, and transcriptomic measurement conditions.

These findings suggest that mortality-associated transcriptomic structure is partially conserved across independent sepsis cohorts while also highlighting the opportunity for future multimodal models that combine molecular, physiological, and clinical predictors.

4 Statistical and Distributional Analysis

4.1 Batch-Effect Validation

After ComBat harmonization, the atlas was checked to ensure that cohort-specific technical variation was reduced while mortality-associated biological variation remained detectable. This step was important because batch correction should align datasets without flattening the outcome signal needed for prognostic modeling.

4.1.1 PCA-Based Variance Summary

Principal component analysis (PCA) was used to summarize the 7,964-gene expression space into a small number of orthogonal axes. For a standardized expression matrix X , PCA estimates the covariance matrix C and solves:

$$CW = \lambda W$$

The first three principal components (PC1–PC3) were used as compact coordinates for variance testing. These components captured broad transcriptomic structure while making it easier to evaluate whether samples grouped by cohort or by clinical outcome.

4.1.2 Residual Cohort Effects

Residual technical variation was assessed by applying one-way ANOVA to PC1, PC2, and PC3 with cohort identity as the grouping variable. The F-statistic compared between-cohort and within-cohort variation:

$$F = \frac{\text{MSB}}{\text{MSW}}$$

where MSB is the mean square between cohorts and MSW is the mean square within cohorts. A large significant value would suggest that samples still cluster by dataset, whereas a small non-significant value would indicate that cohort labels explain little of the remaining PCA structure.

4.1.3 Mortality Signal Preservation

The same ANOVA approach was repeated using mortality status as the grouping variable. This complementary test checked whether survivor and non-survivor samples still showed measurable separation after harmonization.

Together, these analyses evaluated both sides of the harmonization goal: technical cohort effects should be minimized, but clinically relevant mortality signal should remain present for downstream modeling.

4.2 Model Evaluation, Calibration, and Generalizability

After selecting the final 36-gene XGBoost model, four complementary evaluation steps were used to assess whether the signature was statistically meaningful, clinically interpretable, and robust across patient subgroups and cohorts. These analyses examined biomarker distributions, probability calibration, subgroup fairness, and portability across independent datasets.

4.2.1 Distributional Analysis

Before interpreting model predictions, the expression profiles of the 36 selected biomarkers were compared between survivors and non-survivors. This step tested whether the selected genes showed distinct distributional patterns between clinical outcome groups, rather than differing only by small changes in average expression.

The two-sample Kolmogorov-Smirnov (KS) test was used because it is non-parametric and compares the full empirical cumulative distribution functions of two groups. For a given gene, let $F_{ns}(x)$ denote the cumulative distribution function for non-survivors and $F_s(x)$ denote the cumulative distribution function for survivors. The KS statistic is defined as:

$$D = \max_x |F_{ns}(x) - F_s(x)|$$

A larger D value indicates stronger separation between the two expression distributions. When paired with a significant p-value, this suggests that the gene’s expression pattern differs meaningfully between outcome trajectories, not merely at a single summary statistic such as the mean.

4.2.2 Model Calibration

AUROC measures whether the model ranks patients correctly by risk, but it does not determine whether the predicted probabilities are numerically reliable. For example, if a patient receives a predicted mortality probability of 0.80, a well-calibrated model should mean that approximately 80% of patients assigned similar risk actually experience the event.

Calibration was evaluated with the Brier score, which measures the mean squared error between predicted probabilities and observed binary outcomes. For N patients, predicted probability p_i , and observed outcome $o_i \in \{0, 1\}$, the Brier score is:

$$BS = \frac{1}{N} \sum_{i=1}^N (p_i - o_i)^2$$

Lower Brier scores indicate better calibration. In practical terms, this means the model’s predicted risks align more closely with observed clinical event rates, rather than only producing a correct ordering of high- and low-risk patients.

4.2.3 Subgroup Fairness

To evaluate whether the model performed consistently across available demographic subgroups, predictions were stratified by metadata such as biological sex and age group. AUROC was then calculated separately within each subgroup when sufficient samples were available.

Fairness was assessed by comparing predictive discrimination across complementary groups. For example, if g_1 and g_2 represent two demographic subgroups, the absolute AUROC difference can be written as:

$$\Delta\text{AUC} = |\text{AUC}_{g_1} - \text{AUC}_{g_2}| \leq \epsilon$$

Here, $\epsilon = 0.05$ represents the predefined tolerance threshold for acceptable subgroup variation, corresponding to an absolute AUROC difference of no more than five percentage points between demographic groups. This analysis helps identify whether the model is disproportionately accurate for one subgroup while performing worse for another, which is important before considering clinical deployment.

4.2.4 Leave-One-Cohort-Out Generalizability

Standard cross-validation can overestimate portability because patients from the same cohort may appear in both training and validation folds. This can mask dataset shift caused by differences in hospital protocols, sequencing platforms, geography, or patient recruitment. Leave-one-cohort-out (LOCO) validation addresses this by training the model on $K - 1$ cohorts and testing it on the entirely held-out K^{th} cohort.

To summarize how much performance varied across cohorts, heterogeneity was quantified using Cochran’s Q and the I^2 statistic. Given cohort-specific AUROC estimates T_i , variances V_i , precision weights $w_i = 1/V_i$, and pooled performance estimate $\bar{T}$, Cochran’s Q is:

$$Q = \sum_{i=1}^K w_i (T_i - \bar{T})^2$$

The I^2 statistic expresses the proportion of total performance variability attributable to true inter-cohort heterogeneity rather than random sampling error:

$$I^2 = \max\left(0, \frac{Q - df}{Q}\right) \times 100\%$$

where $df = K - 1$. Together, LOCO validation and heterogeneity analysis provide a stricter assessment of how resilient the 36-gene signature is when transferred across independent clinical cohorts.